MalariAI: Automated Malaria Cell Segmentation from Blood Smear Images Using Computer Vision and Deep Learning Techniques.

Kaysarul Anas Apurba, Humayun Ahmad, Tanvir Sadat, Mathew Shawon Rodrick

Abstract: Malaria, a life-threatening infectious disease caused by the Plasmodium parasite, continues to be a significant worldwide health issue, especially in areas with poor access to medical treatment resources. Traditional manual methods of malaria cell segmentation from blood smear images suffer from limitations such as subjectivity, time inefficiency, and reliance on skilled personnel. In response to this critical issue, we introduce "MalariAI," an automated malaria cell segmentation system that leverages state-of-the-art computer vision and deep learning techniques. Our goal is to outline the development and evaluation of MalariAI, highlighting its potential to revolutionize the field of malaria diagnosis. By combining image processing and deep learning, MalariAI achieves remarkable accuracy in identifying and segmenting malaria-infected red blood cells, paving the way for efficient and reliable disease detection. Our approach significantly reduces the burden on healthcare professionals, accelerates diagnosis, and improves diagnostic accuracy. We present the results of extensive experiments, demonstrating the efficacy of MalariAI compared to traditional methods. The findings underscore the system's potential to enhance healthcare outcomes and support malaria control efforts.

Keywords— Malaria. Automated Cell Segmentation. Computer Vision. Image Processing. Healthcare.

I. INTRODUCTION

Caused by the plasmodium parasite, it represents a pervasive global health challenge, particularly in regions with limited access to healthcare infrastructure and resources. It is responsible for millions of infections and hundreds of thousands of deaths annually, predominantly among vulnerable populations, including children and pregnant women. Due to its excellent sensitivity and specificity, microscopic analysis of blood smears for the presence of malaria parasites has been the cornerstone of malaria diagnosis for more than a century. Thin blood smears help identify the kind of parasite causing the infection, whereas thick blood smears help detect the presence of parasites (Centers for Disease Control and Prevention. CDC twenty four seven. Saving Lives, Protecting People, n.d.). The accuracy of the diagnosis is highly dependent on human skill. The liability imposed by widespread diagnoses in areas where disease is endemic, and resources are few might have a negative impact (Michael Hawkes 1, 2009). Alternative methods, including polymerase chain reaction (PCR) and rapid diagnostic tests (RDT), are employed; however, the performance of PCR analysis is constrained (Carl Maximilian Hommelsheim, 2014), and RDTs are less economical in areas where disease is endemic (Michael Hawkes 1, 2009). However, this (Carl Maximilian Hommelsheim, 2014) manual approach needs to be revised. It depends heavily on microscopists' skill and experience, leading to inter- and intra-observer variability. Furthermore, the process is labor-intensive and time-consuming, often resulting in delayed treatment, particularly in remote and resource-constrained areas. In recent years, the intersection of computer vision and deep learning has revolutionized medical image analysis. Leveraging the power of these cutting-edge technologies, automated systems for identifying and segmenting malaria-infected red blood cells from microscopic images have emerged as a promising solution. Our paper presents "MalariAI," an innovative approach to automated malaria cell segmentation from blood smear images. This intelligent system aims to address the challenges inherent in traditional diagnostic methods while delivering a cost-effective and highly accurate diagnostic tool to assist healthcare professionals, especially in regions where resources are scarce.

We will showcase its potential to expedite the detection of malaria parasites and serve as a critical tool for healthcare practitioners in resource-limited regions. Beyond the immediate implications for malaria diagnosis, we explore the broader significance of our research, shedding light on the role of artificial intelligence in medical image analysis and its potential to transform the landscape of infectious disease diagnosis.

II. PRIOR AND RELATED WORKS

Malaria is a widespread and potentially life-threatening disease transmitted through mosquito bites. Early and accurate diagnosis is essential for

effective treatment and preventing the spread of the disease Considering the challenges, researchers have been exploring innovative ways to improve the diagnostic process, particularly using technology.

One of the papers we're reading proposes a fascinating method for solving this issue. In presents " Deep Malaria Parasite Detection in Thin Blood Smear Microscopic Images (Images", 2021)" the authors offer a sophisticated system that uses these methods. This system performs several operations, starting with the picture capture of blood samples and ending with the detection of infected cells and their classification using a support vector machine (SVM). The results are impressive with a sensitivity rate of 93.12% and a specificity rate of 93.17%. This indicates that it is quite good at differentiating between infected and healthy cells. This method is particularly intriguing since it lessens the effort for human diagnosticians and lowers the likelihood of mistakes. This is significant, especially in areas with a shortage of competent workers due to inadequate healthcare resources. So, it's a technological success and could revolutionize malaria diagnosis.

In a different publication, " Computational Methods for Automated Analysis of Malaria Parasite Using Blood Smear Images: Recent Advances" (Shambhu, 2022) the authors focus on deep learning. They discuss how it can be difficult to handle huge microscope images, mainly when using conventional neural networks. Because convolutional neural networks (CNNs) are effective at lightening the computational load, they claim they can be a solution. The study describes CNN's deep-learning model for malaria detection and discusses and contrasts several CNN architectures. This model includes processes for feature extraction, classification, and data preprocessing. It scores better on the NIH malaria dataset than other deep-learning models, which is interesting. It achieves high assessment criteria for accuracy, sensitivity, and specificity. Additionally, it saves time during training and testing, which is essential for practical application.

The study "U-Net: Convolutional Networks for Biomedical Image Segmentation" (Ronneberger, 2015) offers a deep learning architecture built for precise and efficient biomedical image segmentation. The U-Net architecture is specifically designed to address the special issues of medical image processing, where correct identification of anatomical features or anomalies is crucial. The U-Net model has an encoder-decoder structure that is symmetric. The encoder part uses convolutional layers to gather contextual information, while the decoder part generates a high-resolution segmentation map. U-Net's skip connections, which allow it to mix low-level and high-level information, preserve fine-grained details during the up-sampling process, make it highly effective. This method has been used in a variety of medical image segmentation tasks, such as the delineation of tumors, blood arteries, and cell nuclei. The paper's uniqueness is from its capacity to generate very accurate and thorough segmentations, making it a helpful tool for medical practitioners and researchers dealing with biological pictures. With multiple extensions and adaptations for specific applications, U-Net has become a core architecture in the field of medical image processing.

III. METHODOLOGY

3.1. System Diagram:

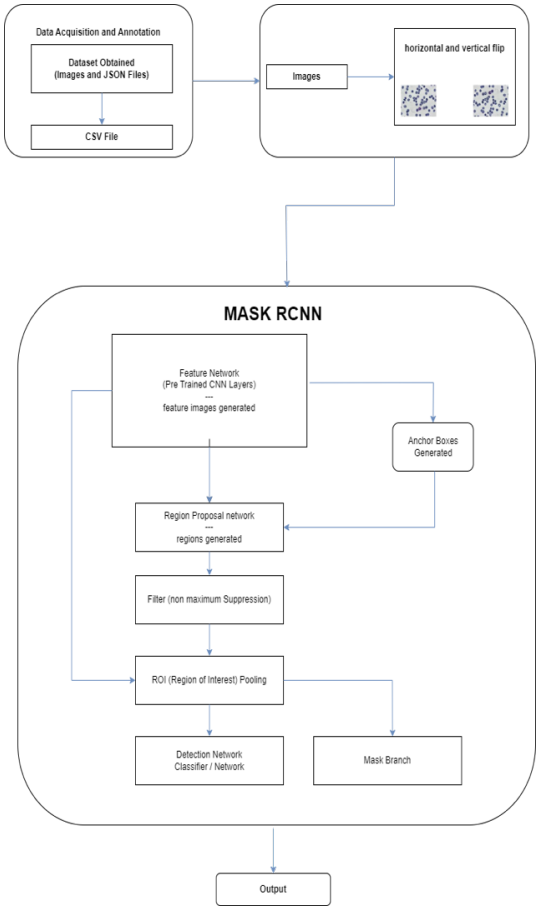

Figure 1 System Diagram

At first, we acquired a suitable dataset. Then we converted the JSON files from the dataset to csv files. Then we applied vertical and horizontal flip to the

images to increase the training size. After that we applied the Mask R-CNN model which has three layers namely the Feature network, RPN layer and Detection Network.

3.2. Dataset: Our study utilized the Malaria Bounding Boxes dataset obtained from Kaggle (al., n.d.). All the images were provided by J. Huang of MIT and the Broad Institute in Cambridge, MA.

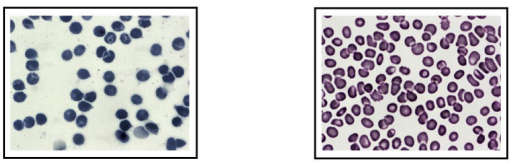

Figure 2 Sample Images from Dataset

The dataset consists of six class labels. Among them two are uninfected and four are infected. These are –

- Uninfected
    1. RBCs
    2. Leukocytes
- Infected
    1. Gametocytes
    2. Rings
    3. Trophozoites
    4. Schizonts

The dataset is hugely imbalanced towards RBC class as 97% of the cells are annotated as RBC. The annotators labeled some of the cells as 'difficult' as they couldn't be identified. The format of the images is in jpg or png. There are around 1364 images (~80000 cells) in the dataset and the images were split into the ratio of 8:2 for training and testing purposes respectively by us (al., n.d.).

3.3. Data Preprocessing:

3.3.1. Data Annotation:

The bounding box information of the images were provided in JSON files. So, for our work, we converted it to a .csv file. The .csv file contained information about each bounding box with path information of the image, minimum and maximum for both x- axis and y-axis of the bounding boxes. The file also included a class label for each box.

image	y_max	y_min	x_max	x_min	class_label
/images/8d02117d-6	1158	1057	1540	1440	red blood cell
/images/8d02117d-6	971	868	1403	1303	red blood cell
/images/8d02117d-6	689	578	1008	900	red blood cell
/images/8d02117d-6	408	304	713	611	red blood cell
/images/8d02117d-6	312	198	1003	881	red blood cell
/images/8d02117d-6	293	193	1574	1480	red blood cell
/images/8d02117d-6	364	257	1502	1384	red blood cell
/images/8d02117d-6	533	430	392	302	red blood cell
/images/8d02117d-6	863	747	1308	1191	red blood cell
/images/8d02117d-6	1087	992	552	454	red blood cell

Figure 3 CSV Format Data

From the .csv file, we created an annotation file which we used for feeding data into our model.

```
H:\\Cell Segmentation\\4th try\\CanWeDoThis\\images\\8d02117d-6c71-4e47-b50a-
6cc8d5eb1d55.png,1440,1057,1540,1158,red blood cell
H:\\Cell Segmentation\\4th try\\CanWeDoThis\\images\\8d02117d-6c71-4e47-b50a-
6cc8d5eb1d55.png,1303,868,1403,971,red blood cell
H:\\Cell Segmentation\\4th try\\CanWeDoThis\\images\\8d02117d-6c71-4e47-b50a-
6cc8d5eb1d55.png,900,578,1008,689,red blood cell
H:\\Cell Segmentation\\4th try\\CanWeDoThis\\images\\8d02117d-6c71-4e47-b50a-
6cc8d5eb1d55.png,611,304,713,408,red blood cell
H:\\Cell Segmentation\\4th try\\CanWeDoThis\\images\\8d02117d-6c71-4e47-b50a-
6cc8d5eb1d55.png,881,198,1003,312,red blood cell
H:\\Cell Segmentation\\4th try\\CanWeDoThis\\images\\8d02117d-6c71-4e47-b50a-
6cc8d5eb1d55.png,1480,193,1574,293,red blood cell
H:\\Cell Segmentation\\4th try\\CanWeDoThis\\images\\8d02117d-6c71-4e47-b50a-
6cc8d5eb1d55.png,1384,257,1502,364,red blood cell
H:\\Cell Segmentation\\4th try\\CanWeDoThis\\images\\8d02117d-6c71-4e47-b50a-
6cc8d5eb1d55.png,302,430,392,533,red blood cell
H:\\Cell Segmentation\\4th try\\CanWeDoThis\\images\\8d02117d-6c71-4e47-b50a-
6cc8d5eb1d55.png,1191,747,1308,863,red blood cell
H:\\Cell Segmentation\\4th try\\CanWeDoThis\\images\\8d02117d-6c71-4e47-b50a-
6cc8d5eb1d55.png,454,992,552,1087,red blood cell
```

Figure 4 Data Annotation on CSV File

3.3.3 Data Augmentation:

For data augmentation purpose, we flipped each image vertically and horizontally. The bounding boxes were adjusted accordingly for each flipped image. Due to this, our dataset had more images for training the model.

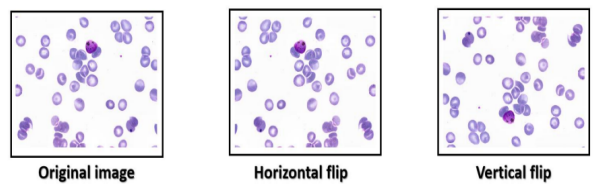

Figure 5 Augmented Images from an Original Image of Dataset

3.4. Model:

Our deep learning model was the Mask R-CNN model. It is a popular model for real time object detection. It works mainly based on three neural networks (S. Goswami, 2018). The first one is the 'Feature Network'. This one works as a pre-trained image classifier. It tries to extract all the important features from the images while, maintaining size and shape of

the original input images. This model extracts 32*32*3 feature maps and it has 64 filters that are being executed in four blocks. After these, total params are 458,257. 442,721 are trainable params. 15,576 are non-trainable params. In the following figure the first 3 stages of the feature network acts as the base of the neural network and the fourth or the last stage is the classifier.

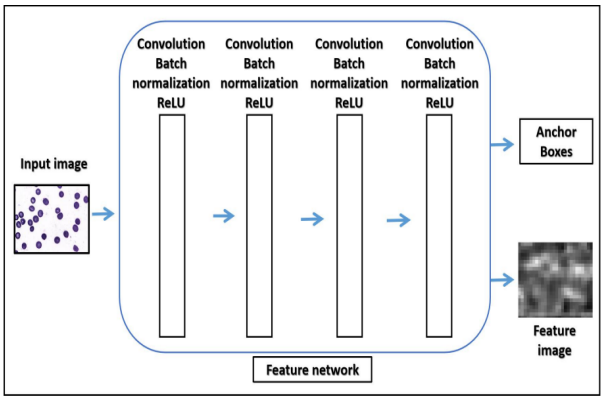

Figure 6 Feature Network of Mask R-CNN

The train images we used have colors and varies in sizes. They are fed into the network as 3 separate matrices. The 3 matrices hold the RGB values of each pixel. These images are represented as (height * width * depth) tensors (multidimensional arrays) which are passed through the pre-trained CNN up until an intermediate layer. This gives a convolutional feature map that is used as a feature extractor for the next part. Then comes the RPN or 'Regional Proposal Network'. This neural network tries to find out which places of an image contain objects. It creates many 'Region of Interests' and represents them as bounding boxes.

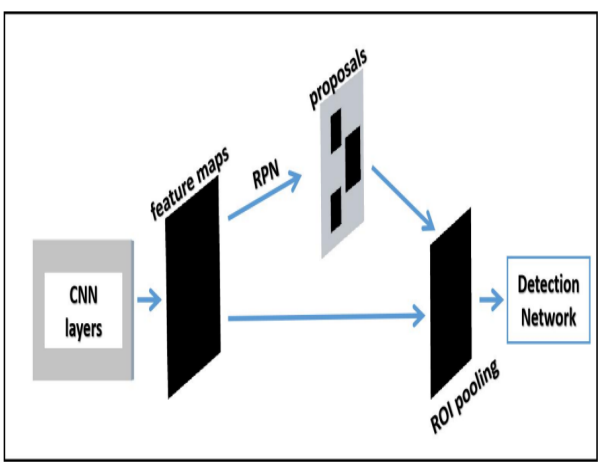

Figure 7 RPN network of Mask R-CNN

The output of the feature network is a feature image, and every pixel of that image is considered as an anchor. Anchors are fixed size reference bounding boxes that are used by the RPN (Rey, 2018). RPN predicts the possibility of an anchor being an object.

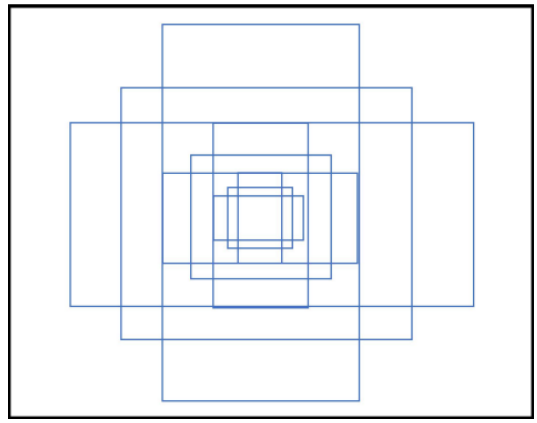

Figure 9. Anchor Boxes for a single position

The model does not locate the objects in the image instead it determines which of these bounding boxes are more likely to contain an object. NMs or non-maximum suppression is used to remove the boxes that overlap with other boxes that have higher scores (Gao, 2017; Rey, 2018). Using the list of the possible candidates or the regions of interest (ROI) and the features extracted from the CNN we apply ROI pooling. Intersection over union (IOU) over all bounding boxes with the ground truth boxes are taken. RPN also tries to tighten the center of the bounding boxes. The final network is the 'Detection network' and the Mask Branch. It takes the output of the previous two layers and then decides on the final bounding boxes for objects (Gao, 2017). Much like the RPN network IOU is taken over all the ground truth boxes are taken. The IOU value for our model is 0.5. The IOU labels the ROIs as foreground if the value is greater than 0.5 and otherwise it is labeled as background. The pool size is 7 meaning the features are cropped and max-pooled to 7*7 size before entering the Detection Network and Mask Branch as a batch. For classification, it is needed to train both detection network and RPN separately. However, for our work, which is only cell segmentation, we didn't need to train them separately.

3.5. Result Analysis:

After training the Mask R-CNN model, we tested some images and checked our results. We split the

dataset into train and test images in 8:2 ratio. These sample images were taken from the test images.

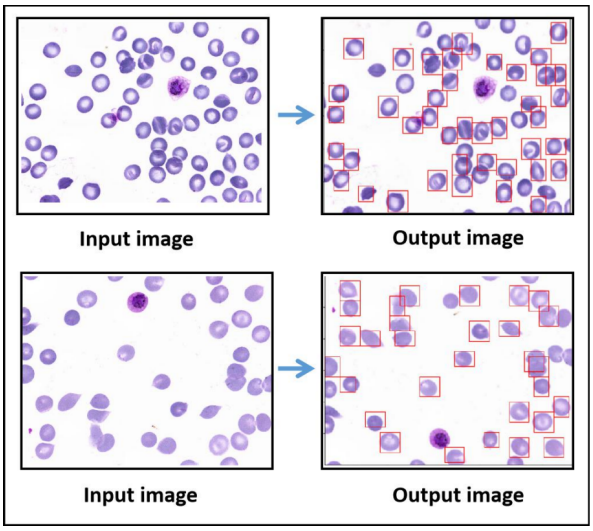

Figure 8 Results of sample test Images.

Our model takes approximately 7-10 seconds to test each image. To assess accuracy, we employed two main methods. Firstly, we compared generated bounding boxes with ground truth boxes for sample test images, achieving an 86% accuracy. Secondly, we manually counted the accuracy for 50 test images by comparing the original number of cells with those recognized by our model, resulting in a 77% accuracy.

Our analysis revealed a challenge in properly identifying cells near image edges, possibly due to inadequate annotation in the dataset. Many ground truth boxes did not include cells at image edges, causing a disparity between accuracy calculated based on ground truth boxes and manual counting.

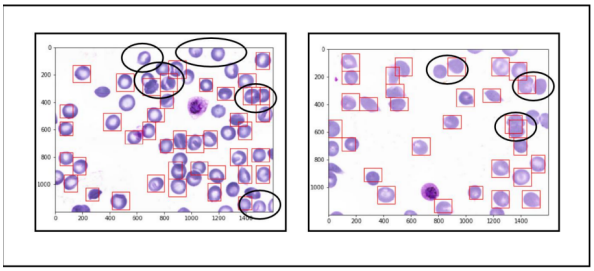

Figure 9 Images with missing Bounding Boxes at edges and detected Overlapping Cells

IV. Conclusions

Segmenting cells is an important task in the medical field. For detecting various diseases, we need to perform the segmentation. And doing this by manual labour can be very time consuming. Our work uses the deep learning methodology to segment cells using the Mask R-CNN model.

Although we have achieved some level of success, but we would like to properly annotate the dataset as many cells in the images were missing bounding boxes. This might help us to increase our accuracy. Also, our work only includes the part of segmenting cells from microscopic images, but these segmented images can further be used to detect the presence of parasitic cells. Using a deep learning model, we can classify the cell images and distinguish the infected and the uninfected ones.

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